

ident, **x**, **y**, **this**
location, *ℓ*
class_name, **C**, **D**, **E**
field_name, **f**
method_name, **m**
constr_name, **K**
index, *h*, *i*, *j*, *k*, *n*

<i>class_def</i>	$::=$ $ \quad \mathbf{class\,C\,extends\,D}\{C_1\,f_1, \dots, C_i\,f_i; K, M_1, \dots, M_j\}$	
<i>constr_def, KK</i>	$::=$ $ \quad C(C_1\,x_1, \dots, C_i\,x_i, C'_1\,x'_1, \dots, C'_j\,x'_j)\{\mathbf{super}\,(x_1, \dots, x_k); f_1 = x'_1, \dots, f_h = x'_h; \}$	
<i>method_def, M</i>	$::=$ $ \quad C\mathbf{m}(C_1\,x_1, \dots, C_i\,x_i)\{\mathbf{return\,e}; \}$	
<i>expr, e</i>	$::=$ $ \quad \mathbf{x}$ $ \quad \mathbf{e.m(e_1, \dots, e_n)}$ $ \quad \mathbf{e.f}$ $ \quad \mathbf{new\,C(e_1, \dots, e_n)}$ $ \quad (\mathbf{C})\mathbf{e}$ $ \quad (\mathbf{e})$ $ \quad \{\mathbf{e_1/x_1, \dots, e_k/x_k, this/e'}\mathbf{e''}$	M M
<i>terminals</i>	$::=$ $ \quad \longrightarrow$ $ \quad \rightarrow$ $ \quad \mathbf{extends}$ $ \quad \mathbf{return}$ $ \quad =$ $ \quad <:$ $ \quad \nlessdot:$ $ \quad \neq$ $ \quad \vdash$	
Γ	$::=$ $ \quad \mathbf{x : C}$ $ \quad \Gamma, \Gamma'$ $ \quad \Gamma_1, \dots, \Gamma_n$	M M
<i>formula</i>	$::=$ $ \quad \mathbf{judgement}$ $ \quad \mathbf{formula_1 \dots formula_n}$ $ \quad \mathbf{C \neq D}$ $ \quad \mathbf{C \nlessdot D}$ $ \quad \mathbf{fields\,(C) = C_1\,f_1, \dots, C_n\,f_n}$ $ \quad \mathbf{stupid}$ $ \quad \mathbf{CT\,(C) = class_def}$ $ \quad \Gamma = \Gamma'$ $ \quad K = KK$ $ \quad \mathbf{mbody\,(m, C) = (x_1, \dots, x_n, e)}$ $ \quad 1 \leq i \leq n$	
<i>Jop</i>	$::=$ $ \quad \mathbf{x : C \in \Gamma}$	

		mtype ($\mathbf{m}, \mathbf{C}$) = $\mathbf{D}_1, \dots, \mathbf{D}_n \rightarrow \mathbf{C}'$
		override ($\mathbf{C}_1, \dots, \mathbf{C}_n \rightarrow \mathbf{C}', \mathbf{m}, \mathbf{D}$)
		$\mathbf{C} < : \mathbf{D}$
		$\mathbf{e} \longrightarrow \mathbf{e}'$
		$\Gamma \vdash \mathbf{e} : \mathbf{C}$
		<i>class_def</i> OK
		<i>M</i> OKIN $\mathbf{C}$

<i>judgement</i>	$::=$
	<i>Jop</i>

<i>user_syntax</i>	$::=$
	<i>ident</i>
	<i>location</i>
	<i>class_name</i>
	<i>field_name</i>
	<i>method_name</i>
	<i>constr_name</i>
	<i>index</i>
	<i>class_def</i>
	<i>constr_def</i>
	<i>method_def</i>
	<i>expr</i>
	<i>terminals</i>
	Γ
	<i>formula</i>

$\mathbf{x} : \mathbf{C} \in \Gamma$

$\overline{\mathbf{x} : \mathbf{C} \in \Gamma}$	TIN_1
$\overline{\mathbf{x} : \mathbf{C} \in \Gamma}$	
$\mathbf{x} : \mathbf{C} \in \Gamma, \mathbf{x}' : \mathbf{C}'$	TIN_2

$\mathbf{mtype}(\mathbf{m}, \mathbf{C}) = \mathbf{D}_1, \dots, \mathbf{D}_n \rightarrow \mathbf{C}'$

$\overline{\mathbf{mtype}(\mathbf{m}, \mathbf{C}) = \mathbf{D}_1, \dots, \mathbf{D}_n \rightarrow \mathbf{C}}$	GET_PLACEHOLDERDEFINTION
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$\mathbf{override}(\mathbf{C}_1, \dots, \mathbf{C}_n \rightarrow \mathbf{C}', \mathbf{m}, \mathbf{D})$

$\overline{\mathbf{override}(\mathbf{C}_1, \dots, \mathbf{C}_n \rightarrow \mathbf{C}_0, \mathbf{m}, \mathbf{D})}$	OVR_PLACEHOLDERDEFINTION
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$\mathbf{C} < : \mathbf{D}$

$\overline{\mathbf{C} < : \mathbf{C}}$	S_REFL
$\mathbf{C} < : \mathbf{D}$	
$\mathbf{D} < : \mathbf{E}$	
$\mathbf{C} < : \mathbf{E}$	S_TRANS

$\mathbf{CT}(\mathbf{C}) = \mathbf{class} \mathbf{C} \mathbf{extends} \mathbf{D} \{ \mathbf{C}_1 \mathbf{f}_1, \dots, \mathbf{C}_i \mathbf{f}_i; \mathbf{K}, \mathbf{M}_1, \dots, \mathbf{M}_j \}$	
$\ll \text{no parses (char 2): } \mathbf{c} \mathbf{***<: D} \gg$	S_INH

$e \longrightarrow e'$

$$\begin{array}{c}
\frac{1 \leq i \leq k \quad \mathbf{fields}(C) = C_1 f_1, \dots, C_n f_n}{\ll \text{no parses (char 2): :d***eeper: (new C(e1 , \dots, en)) .fi --> ei} \gg} \text{R_FIELD} \\
\frac{\mathbf{mbody}(m, C) = (x_1, \dots, x_k, e)}{\ll \text{no parses (char 2): :d***eeper: (new C(e1 , \dots, en)).m(e'1, \dots, e'k) --> \{e'1/x1, \dots, e'k/xk, t\}} \gg} \\
\frac{C <: D}{\ll \text{no parses (char 2): :d***eeper: (D)(new C(e1, \dots, en)) --> new C(e1, \dots, en)} \gg} \text{R_CAST}
\end{array}$$

$\Gamma \vdash e : C$

$$\begin{array}{c}
\frac{x : C \in \Gamma}{\Gamma \vdash x : C} \text{TY_VAR} \\
\frac{\Gamma \vdash e_0 : C_0 \quad \mathbf{mtype}(m, C_0) = D_1, \dots, D_n \rightarrow C \quad \Gamma \vdash e_1 : C_1 \quad \dots \quad \Gamma \vdash e_n : C_n \quad C_1 <: D_1 \quad \dots \quad C_n <: D_n}{\Gamma \vdash e_0.m(e_1, \dots, e_n) : C} \text{TY_INVK} \\
\frac{\mathbf{fields}(C) = D_1 f_1, \dots, D_n f_n \quad \Gamma \vdash e_1 : C_1 \quad \dots \quad \Gamma \vdash e_n : C_n \quad C_1 <: D_1 \quad \dots \quad C_n <: D_n}{\Gamma \vdash \mathbf{new} C(e_1, \dots, e_n) : C} \text{TY_NEW} \\
\frac{\Gamma \vdash e : D \quad D <: C}{\Gamma \vdash (C)e : C} \text{TY_CAST1} \\
\frac{\Gamma \vdash e : D \quad C <: D \quad C \neq D}{\Gamma \vdash (C)e : C} \text{TY_CAST2} \\
\frac{\Gamma \vdash e : D \quad C \not<: D \quad D \not<: C \quad \mathbf{stupid}}{\Gamma \vdash (C)e : C} \text{TY_CAST3}
\end{array}$$

class_def OK

$$\frac{K = C(D_1 x_1, \dots, D_i x_i, C_1 y_1, \dots, C_j y_j) \{ \mathbf{super}(x_1, \dots, x_i); f_1 = y_1, \dots, f_j = y_j; \} \quad \mathbf{fields}(D) = D_1 f'_1, \dots, D_i f'_i \quad M_1 \text{OKIN } C \quad \dots \quad M_n \text{OKIN } C}{\mathbf{class} C \text{ extends } D \{ C_1 f_1, \dots, C_j f_j; K, M_1, \dots, M_n \} \text{OK}} \text{MCL_1}$$

$M \text{OKIN } C$

$$\frac{\begin{array}{l} x_1 : C_1, \dots, x_n : C_n \vdash e' : D' \\ D' <: C' \\ \mathbf{CT}(C) = \mathbf{class} C \text{ extends } D \{ C'_1 f_1, \dots, C'_i f_i; K, M_1, \dots, M_j \} \\ \mathbf{override}(C_1, \dots, C_n \rightarrow C', m, D) \end{array}}{C' m(C_1 x_1, \dots, C_n x_n) \{ \mathbf{return} e'; \} \text{OKIN } C} \text{MTY_METH}$$

Definition rules:	14 good	4 bad
Definition rule clauses:	46 good	4 bad